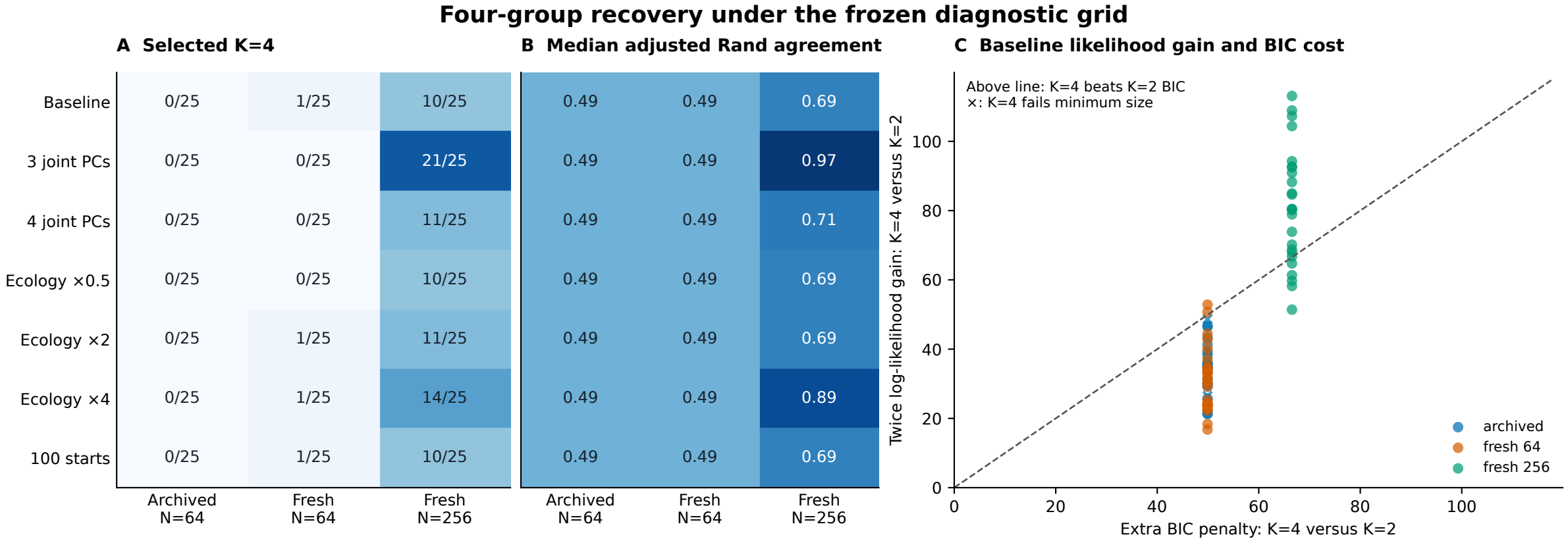
Four-group recovery under the frozen diagnostic grid
A Selected K=4
B Median adjusted Rand agreement
C Baseline likelihood gain and BIC cost
Baseline
3 joint PCs
4 joint PCs
Ecology ×0.5
Ecology ×2
Ecology ×4
100 starts
0/25
1/25
10/25
0/25
0/25
21/25
0/25
0/25
11/25
0/25
0/25
10/25
0/25
1/25
11/25
0/25
1/25
14/25
0/25
1/25
10/25
0.49
0.49
0.69
0.49
0.49
0.97
0.49
0.49
0.71
0.49
0.49
0.69
0.49
0.49
0.69
0.49
0.49
0.89
0.49
0.49
0.69
Archived N=64
Fresh N=64
Fresh N=256
Archived N=64
Fresh N=64
Fresh N=256
Above line: K=4 beats K=2 BIC
×: K=4 fails minimum size
Twice log-likelihood gain: K=4 versus K=2
Extra BIC penalty: K=4 versus K=2
0
20
40
60
80
100
archived
fresh 64
fresh 256
25 datasets per cohort; configurations remain diagnostic. No bootstrap or significance reruns.